JA

Joshua Alcobia Gomes Mechanical Engineer

MECHANICAL ENGINEER · AUTOMOTIVE & AEROSPACE

Second-year MEng Mechanical Engineering student at UCL (predicted First), heading to McLaren on an industrial placement for 2026–27. I work across automotive and aerospace — from CFD and rocketry to pharmaceutical modelling, machine learning and AI tooling.

DISCIPLINE	UNIVERSITY	EMAIL	LINKEDIN
Mechanical Engineering	UCL · MEng	jg7alcobia@gmail.com	josh-alcobia-gomes

Curiosity, engineered.

I'm a mechanical engineer who has always been fascinated by every incremental gain, whatever I'm working on. The highest-stakes work demands the most attention to detail, and that's exactly where I excel. I finished my first two years at UCL with 83% and 82% (First Class Honours predicted), and I'm now on industrial placement at McLaren for 2026–27, before returning to UCL for third year.

I've worked across consulting, pharmaceutical and automotive, picking up whatever software the job demands. I enjoy leading teams as much as learning from the people around me, and there's always more to pick up.

Away from engineering I compete nationally in track and field, which is where a lot of the discipline behind this work comes from. I also enjoy football, reading, cooking and hiking.

`Fusion 360 · SolidWorks · Onshape · AutoCAD · ANSYS SpaceClaim` `ANSYS Fluent & Mechanical`

`COMSOL · FloEFD · OpenRocket · QBlade` `Python · MATLAB · C++` `CFD & FEA`

`Machine Learning · LLMs · Agentic Workflows` `3D printing · laser cutting`

AT A GLANCE

Discipline: MEng Mechanical Engineering
University: UCL · 3rd year (83% & 82%, First Class Honours predicted)
Current: McLaren Automotive, Industrial Placement (2026–27)
Focus: Aerospace · Automotive · Motorsport · Robotics
Based in: London, UK
Previously: Singapore · Basel
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CONTENTS

A separate 2-page résumé accompanies this portfolio.

McLaren

01 · Automotive / Motorsport

My industrial placement year in high-performance automotive engineering.

ROLE	TEAM	PERIOD	DOMAIN
Industrial Placement	TBC	2026–27 academic year	Automotive · Motorsport

I'm working at McLaren for my industrial placement across the 2026–27 academic year, in a department that's still to be confirmed. It's a full year embedded in one of the sharpest engineering environments in the world, and this page will grow into a record of the projects I work on there.

APPROACH

1. Apply the design, simulation and iterative-process fundamentals built at UCL and NING Research, and keep expanding them.
2. Learn the specific skills high-performance automotive and motorsport demand, and start contributing.
3. Continue developing soft skills.

PROJECTS

3 projects

Placement project 1

COMING SOON

Project details will appear here once I can share them.

Placement project 2

COMING SOON

Project details will appear here once I can share them.

Placement project 3

COMING SOON

Project details will appear here once I can share them.

NING Research

02 · Research & Development

Applied CFD and engineering simulation, turning real client problems into validated models.

ROLE	FIELD	DOMAIN
Simulation / CFD Engineer	CFD & Engineering Simulation	R&D · Consulting

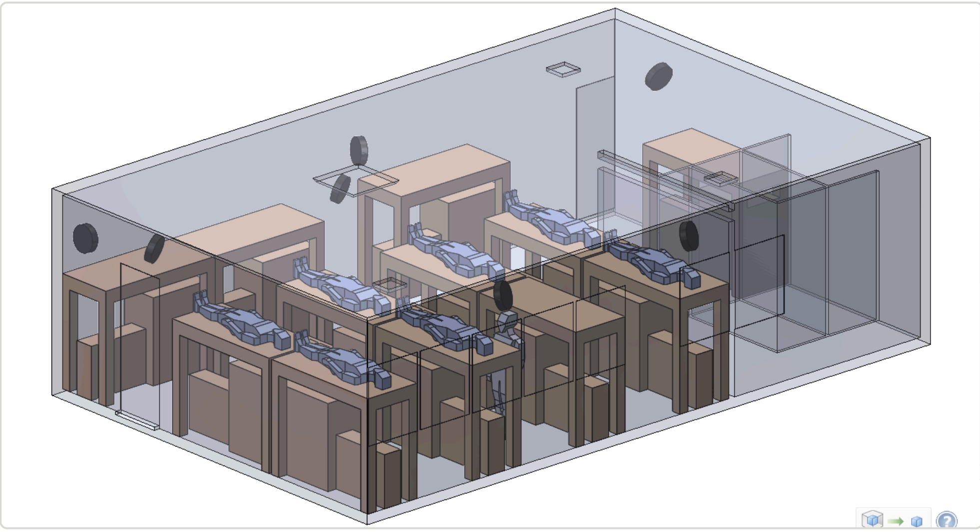
At NING Research I worked on computational fluid dynamics and engineering-simulation projects for a range of clients, from multinational corporations to Olympic medallists.

APPROACH

1. Define the physical problem and success criteria with the client.
2. Build and mesh the CFD / simulation model (OpenFOAM, ANSYS).
3. Validate against data, then translate results into clear recommendations and reports.

PROJECTS

10 projects

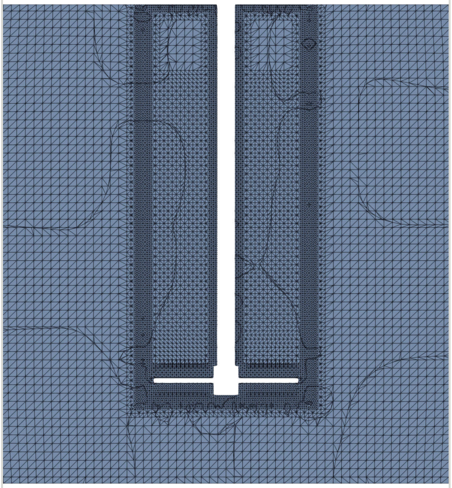


Coughing Dispersal Simulation

CFD · VENTILATION

A Discrete Phase Model study of how cough particles disperse through a room, tested against government compliance standards. Industry: healthcare / built-environment ventilation.

- Varied particle size and the cough velocity profile across cases to cover a realistic range of conditions.
- Wrote a user-defined function (UDF) to track particles through the recirculatory fans.
- Adjusted boundary conditions per case to test compliance under different ventilation setups.

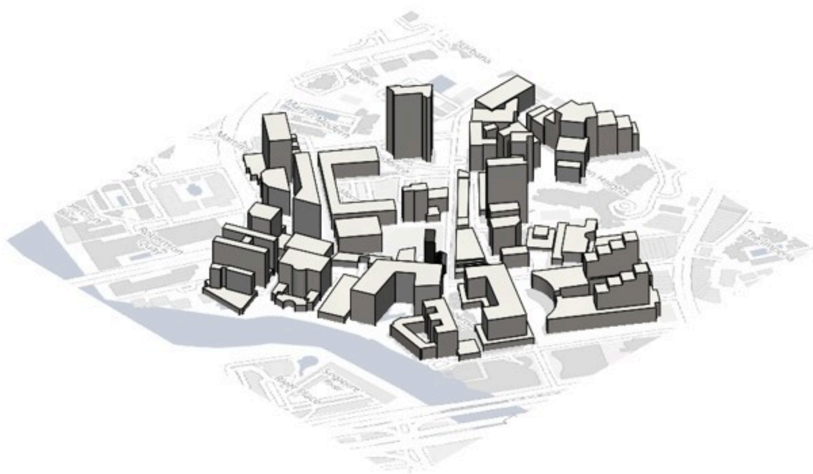


OpenFOAM Mixing Tank

CFD · MIXING

An end-to-end mixing-tank study, from CAD through simulation to a machine-learning model predicting mixing extent. Industry: industrial process engineering.

- Built impeller mixing zones from the CAD model and meshed in OpenFOAM.
- Simulated using interFoam with AMI sliding interfaces for the rotating impeller.
- Post-processed in ParaView, then fed the results into a machine-learning algorithm to predict mixing extent.

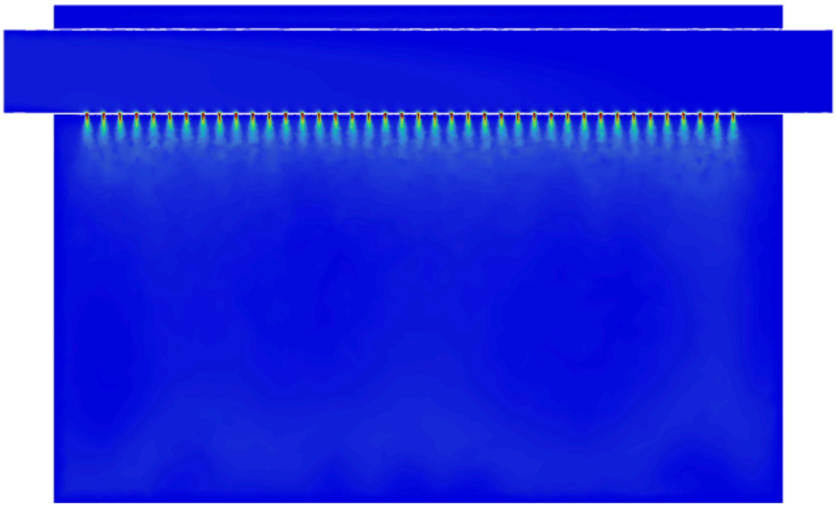


Hotel Rooftop Chiller Ventilation Study

CFD · VENTILATION

A rooftop ventilation study for a hotel adding chiller units so it could expand its room count. Industry: hospitality / built environment, Singapore.

- Tested and orientated the cooling units to eliminate recirculation of exhaust air.
- Determined the efficiency of each layout to confirm the chillers stayed within their performance envelope.
- Verified the final arrangement met the client's expectations before installation.



Sports Hall Thermal Comfort Study

CFD · BUILT ENVIRONMENT

A thermal-comfort comparison for a large sports hall in Toa Payoh, testing two air-delivery methods across two solvers. Industry: sports / built-environment ventilation, Singapore.

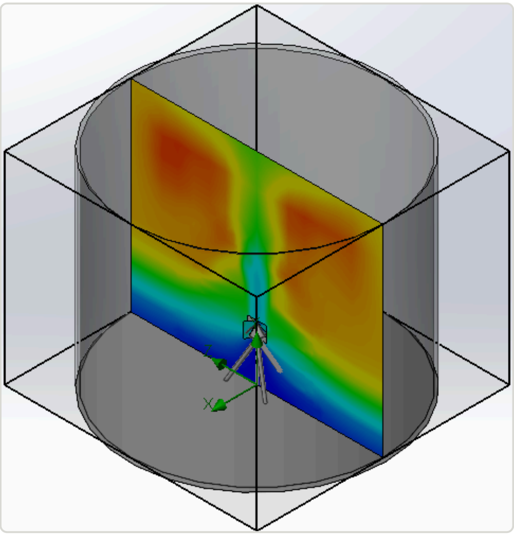
- Compared thermal comfort from small perforated tubing against a porous membrane.
- Ran the same cases in both FloEFD and ANSYS Fluent to compare solver results.

Roof Rainfall Project

CFD · MULTIPHASE

Multiphase CFD study of rainfall on a roof, chaining three solvers to follow water from airborne droplet to standing pool.

- Coupled Discrete Phase Model → Eulerian Wall Film → Volume of Fluid (DPM → EWF → VOF), so falling droplets, the thin film they form on the surface, and bulk pooling water were each handled by the appropriate physics model.
- Built as a feasibility test to prove the handoff between the three models could be made to work end-to-end.

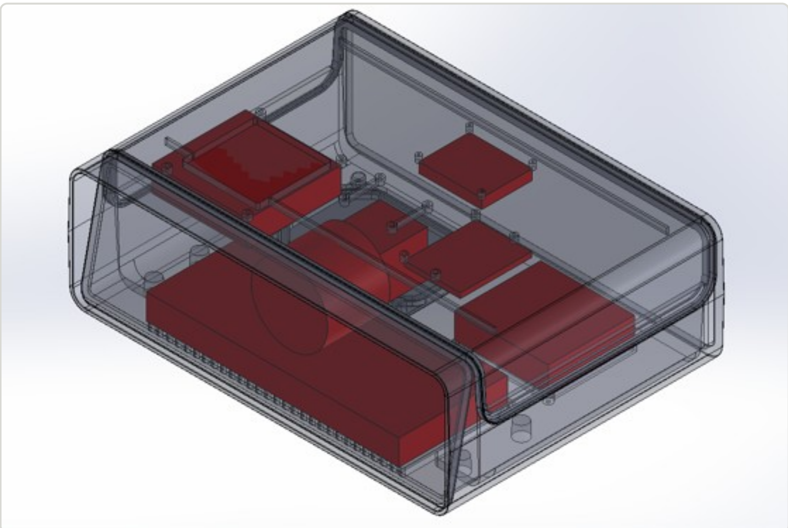


Reservoir Mixer Thermal Stratification Study

CFD · MIXING

A mixer designed to prevent thermal stratification in Singapore's water-supply tanks, evaluated across flow and mixing combinations. Industry: water utilities, Singapore.

- Designed an initial mixer model for installation in the storage tanks.
- Compared three configurations: no flow with mixing, flow without mixing, and flow with mixing.
- Results showed a mixer was not strictly required, though one was still recommended on the basis of supplementary research.



Safety Monitoring Unit Thermal Study

CFD · THERMAL

Internal-component CFD in FloEFD, checking whether the existing thermal management could regulate component temperatures. Industry: construction safety technology.

- Modelled the internal components and their heat loads in FloEFD.
- Assessed whether the current cooling approach held components within their operating temperature range.

KiteFoil Aerodynamics

CFD · SPORTS AERO

Aerodynamic modelling of a kite foil, testing surface geometries and rider positions to reduce drag. Industry: elite sports equipment.

- Modelled the foil and investigated surface geometries to maximise aerodynamic efficiency.
- Reviewed research literature on drag-reduction techniques, then ran CFD studies on each.
- Simulated the athlete's orientation through the race to assess its aerodynamic effect.

Sanitary Pad Glue Application Thermal Study

THERMAL · PROCESS SIMULATION

A thermal study of glue application on a sanitary pad production line, run alongside the FEA team to qualify cheaper materials. Industry: consumer healthcare manufacturing.

- Tested a wide range of process variables to keep production running with alternative materials.
- Worked in conjunction with the FEA team so thermal and structural results lined up.
- Supported a material change aimed at reducing cost.

GENSET and Data Hall CFD

CAD · CFD · THERMAL

CAD and CFD study of a generator set and data hall. Industry: data centre infrastructure.

Novartis

03 · Pharma / Machine Learning

Scientific Modelling Intern — machine learning, simulation and AI tooling in pharmaceutical manufacturing.

ROLE	PERIOD	LOCATION	DOMAIN
Scientific Modelling Intern	**Aug – Sept 2025**	**Basel, Switzerland**	**R&D · Pharma / ML**

Over a summer internship in Basel I built machine-learning models, simulations and AI tools that removed manual work and cut costs across pharmaceutical R&D — spanning powder characterisation, process analytics and LLM-based automation.

APPROACH

1. Turn messy sensor and process data into clean, usable models.
2. Automate repetitive analysis with Python, dashboards and LLM workflows.
3. Validate against real batches and quantify the time and cost saved.

PROJECTS

5 projects

ML Powder Flow Classification

MACHINE LEARNING

Machine-learning models to predict powder flow characteristics from experimental data.

- Used LazyClassifier for rapid model selection, then Grid Search, Lasso regularisation and Bayesian optimisation for hyperparameter tuning.
- Saved CHF 40,000 per batch of APIs by predicting behaviour ahead of manufacture.

MCP Servers for Internal AI Tooling

AI INFRASTRUCTURE

Full write-up in progress.

Streamlit Analytics Dashboards

DATA / PYTHON

Interactive Streamlit applications that eliminated manual data searching for the department.

- Built dropdown-driven apps for dwell-time analysis and screw customisation.
- Developed Python programs for Loss-on-Drying calculations from sensor data, solving mass-balance equations for moisture content and drying efficiency.

AI Project Summariser (LLMs + Agentic Workflows)

LLMS · AGENTS

An AI summariser that gathers data from internal and external databases automatically.

- Used automated workflows and LLMs to compile project context on demand.
- Reduced research time from ~50 minutes to ~15, saving weeks of cumulative effort.

Pill CAD Design for FEA / COMSOL

CAD · SIMULATION

Tablet geometry modelled for high-performance-compute simulation.

- Designed tablets in SolidWorks from technical drawings for COMSOL HPC simulation.
- Saved thousands per batch through predictive modelling.

UCL

04 · Education

MEng Mechanical Engineering — the competition teams and projects that shaped how I engineer.

DEGREE	YEARS	PREDICTED	DOMAIN
MEng Mechanical Engineering	Sept 2024 – June 2028	First Class (83%)	Education

I'm in my second year of an MEng in Mechanical Engineering at UCL, predicted a First (83%). Alongside the degree I throw myself into engineering teams and competitions — rocketry, motorsport, design challenges and more — where the theory meets real hardware.

APPROACH

1. Apply core mechanical engineering: thermofluids, solid mechanics, dynamics, materials.
2. Specialise toward automotive and aerospace through team projects.
3. Lead and deliver — from budgets and Gantt charts to manufacture and test.

PROJECTS

4 projects

UCL Racing — Rocket

AEROSPACE · TEAM LEAD

National Rocketry Championship team lead, aerostructures & simulations engineer.

- Led an 8-engineer team with a £1,000 budget to 8th of 50 at the National Rocketry Championship.
- Validated ANSYS Fluent CFD against drag-coefficient data to 97% accuracy.
- Designed the aerostructure in Fusion 360, increasing apogee by 203 m.
- Ran the project in Jira with Gantt charts, risk assessments and sponsorship outreach.

Formula Student

AUTOMOTIVE · MOTORSPORT

Full write-up in progress.

IMechE Design Challenge

DESIGN & MANUFACTURE

Design & manufacturing engineer — runner-up, Greater London Regional Final.

- Secured National Final qualification as one of only two teams to achieve perfect accuracy.
- Won both the Design Award and the Challenge Competition.
- Used Fusion 360 Generative Design to cut mass by 115.87 g (60%) at a safety factor of 2.
- Produced a full Bill of Materials with data visualisation to identify cost drivers.

UCL Mechathon 2025

TURBINES · COMPETITION

Won the competition (team of 6) with a 76 mm-radius wind-turbine design.

- Achieved 29% efficiency at 3.3 V.
- Used QBlade to analyse aerofoil lift-to-drag ratios and optimal angle of attack (95.8% accurate).
- Developed manufacturing methods across milling, 3D printing and laser cutting.

PRESENTATIONS & COURSEWORK

Drop presentation PowerPoints and coursework PDFs into assets/docs/ and list them here.

Presentation / coursework 1

Presentation / coursework 2

Side Projects

05 · Personal builds

The things I build for the love of it — research, robotics and everything between.

THEMES
Aerospace · Robotics · Research

TOOLS
CAD · Arduino · C++ · 3D print

STATUS
Ongoing

A selection of personal and academic projects I've taken end-to-end — from theoretical research papers to fully built, programmed hardware.

APPROACH

1. Start from a question or a problem worth solving.
2. Model, build and iterate — CAD, electronics, code and test.
3. Document the result properly, challenges and all.

PROJECTS

3 projects

Internal Pressure of Thin-Walled Vessels During Collision

RESEARCH · IMPACT

Theoretical research and mathematical modelling on pressure vs. impact force.

- Modelled the relationship between internal pressure and impact force.
- Wrote a 4,000-word research paper on optimal pressure for impact reduction and its applications to safety systems.

Automated Fruit & Vegetable Cutting Robot

ROBOTICS · AUTOMATION

A kitchen robot designed, built and programmed from scratch.

- Built and programmed the robot with Arduino and C++.
- Modelled and optimised components in CAD for functionality and durability.
- Assembled and iteratively refined the mechanical systems and control algorithms.

Pump Impeller CFD with a Moving Reference Frame

CFD · ROTATING MACHINERY

A self-directed CFD study of a pump impeller, built at NING Research to learn how the moving reference frame method works rather than for a client.

- Set up the rotating zone with a moving reference frame (MRF) to model the spinning impeller.
- Started it after asking about MRF at a workshop, with a coursework problem in mind; I built it to understand the method properly, though it never went into the coursework itself.